Complex Hamiltonian learning and KAM stability for magnetic catheter dynamics in microgravity

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Abstract

Magnetically actuated flexible catheters require a stability analysis connecting distributed mechanics, learned dynamics and operational constraints. Motivated by prospective robotic intervention in space, we develop this connection through complex Hamiltonian neural networks (CHNNs) and Kolmogorov–Arnold–Moser (KAM) theory. A variational rod reduction retains elasticity, magnetic loading, fluid interaction and contact. A complex two-output representation encodes a real Hamiltonian and a Rayleigh dissipation potential, with a decoder satisfying an explicit power balance. For the conservative subsystem, statistical derivative estimates and deterministic polynomial-feature certificates connect nonzero training loss to the norms required by KAM theory. A quartic normal-form calculation establishes local positive-measure families of invariant tori for the actual learned models. A fully explicit annular certificate further specifies an amplitude bound and a torus-measure lower bound. In reproducible two-mode Galerkin experiments, all three training replicates pass the verified loss-to-twist condition at a derivative-noise bound of 5×10^{-5}. Matched complex and real polynomial realizations separate representation from approximation power. Integration, resonance and dissipative-power experiments are supplemented by a magnetically driven cantilever discretization check. Clearance and energy inequalities turn model-error and input-work budgets into sufficient operational constraints. The explicit KAM amplitude bound is extremely conservative and does not certify the simulation amplitudes. Application to a particular catheter requires material, actuation and reduction-error calibration.

Keywords

magnetic catheter robots, complex Hamiltonian neural networks, KAM theory, Galerkin reduction, dissipation, stability certification

1 Introduction

Magnetic actuation offers remote force and torque generation for small flexible robotic devices, including tools intended for minimally invasive intervention (Abbott et al. 2020; Kim et al. 2019). Such a device couples distributed elasticity with magnetic fields, surrounding fluid and intermittent wall interaction. Long-duration space missions motivate the study of this interaction in reduced gravity. Spaceflight-associated fluid redistribution and neuro-ocular changes are documented research concerns (NASA n.d.); they do not establish the incidence of a particular cerebrovascular emergency or the clinical readiness of an orbital intervention. Here, microgravity is a mechanical operating condition and space intervention is the motivating application.

Our central question is whether a learned energy model can support a mathematically justified stability statement for a reduced magnetic catheter. Hamiltonian neural networks (HNNs) impose conservative structure through a scalar energy (Greydanus et al. 2019). Dissipative SymODEN and port-Hamiltonian networks extend energy-based learning to losses and inputs (Zhong et al. 2020; Desai et al. 2021). Complex-valued representations provide coupled amplitude and phase channels (Trabelsi et al. 2018; Zhang et al. 2022). KAM theory supplies persistence of many invariant tori under sufficiently small smooth Hamiltonian perturbations (Pöschel 2001; Bounemoura 2020). Connecting these ideas requires more than a low prediction error: the mechanical coordinates, the learned hypothesis class, the perturbation norm and the stability object must agree.

We retain the complete modelling–learning–stability chain. The reduced physical model supplies the Hamiltonian and dissipation targets. The CHNN approximates those targets with a prescribed decoder. A loss-to-norm estimate bounds the Hamiltonian perturbation on the domain

where KAM applies. Nonlinear normal forms supply the twist absent from a purely harmonic prior. Finally, the conservative invariant structures are related to the full port-driven motion through quantitative clearance and energy bounds. Thus the nonconservative mechanics remains part of the analysis, even though the classical KAM theorem is applied to its conservative subsystem.

The contributions are fourfold. First, we give a dimensionally consistent complex representation of conservative energy and dissipative power that reproduces the energy–dissipation structure of the variationally reduced rod dynamics. Second, we prove loss-to-KAM transfer statements, including a finite-feature certificate computable from the training design matrix. Third, we perform the nonlinear twist calculation for both the target and the fitted CHNN instead of substituting an unrelated polynomial. Fourth, we supply executable numerical experiments and explicit inequalities that translate model error, residual forces and magnetic input work into sufficient operating margins.

2 Distributed mechanics and complex energy representation

2.1 Rod mechanics and a variationally consistent reduction

Following the spatial rod formulation (Simo 1985; Antman 2005), let $s \in [0, L]$ denote reference arclength, $r(s,t) \in \mathbb{R}^3$ the centreline and $D(s,t) \in SO(3)$ its director frame. Define

$$\Gamma = D^{\mathsf{T}} r_s, \quad D_s = D\widehat{\kappa}, \quad D_t = D\widehat{\omega}_b. \tag{1}$$

The hat map is defined by $\widehat{a}b = a \times b$. The extensible, unshearable Kirchhoff specialization has $\Gamma = (0, 0, 1+e)^{\mathsf{T}}$; inextensibility additionally sets $e = 0$. Transverse internal forces are then constraint reactions. They must not be set to zero by substituting zero shear into a shearable constitutive law. In spatial components the balances are

$$\rho A r_{tt} = n_s + f_{\mathrm{mag}} + f_{\mathrm{h}} + f_{\mathrm{c}} + \rho A g, \tag{2}$$

$$\partial_t (D \mathcal{J}_b \omega_b) = m_s + r_s \times n + \tau_{\mathrm{ext}}. \tag{3}$$

For constant body inertia, the left side of the second equation is $D(\mathcal{J}_b \dot{\omega}_b + \omega_b \times \mathcal{J}_b \omega_b)$. With stored energy density $W(\Gamma, \kappa)$, unconstrained components of $D^{\mathsf{T}} n$ and $D^{\mathsf{T}} m$ are its strain derivatives. A fixed base specifies position and orientation; a free tip specifies the appropriate force and moment resultants, including any concentrated magnetic load.

For a body-fixed magnetic moment density $\mu_b(s)$ and a static applied field $B_0(r)$, the magnetic potential is

$$V_{\mathrm{mag}} = -\int_0^L (D\mu_b) \cdot B_0(r)\, ds. \tag{4}$$

Its translational and rotational variations give $f_{\mathrm{mag}} = \nabla_r (\mu \cdot B_0)$ at fixed orientation and $\tau_{\mathrm{mag}} = \mu \times B_0$, where $\mu = D\mu_b$. Thus static magnetic loading belongs to the conservative energy. Field modulation supplies explicit time-dependent work or an input port. A fixed-orientation dipole approximation is equivalent to a consistent Maxwell-stress calculation in its regime of validity; the same load is not added twice.

More explicitly, the magnetic force on a finite material segment can be evaluated on a surrounding vacuum or nonmagnetic control surface Σ using

$$F_{\mathrm{mag}} = \int_\Sigma T_M n_\Sigma \, \mathrm{d}A, \qquad T_M = \mu_0^{-1}(BB^{\mathsf{T}} - \tfrac{1}{2}|B|^2 I). \tag{5}$$

Here B is the total field on the control surface, including the self-consistent magnetic contribution of the material. The associated moment is the integral of the moment arm crossed with $T_M n_\Sigma$. The dipole formulas above are its small-magnet reduction when the applied field varies slowly across the magnetic segment (Abbott et al. 2020).

A modified Morison-type local closure, stated per unit reference length, is

$$f_{\mathrm{h}} = m_f a_f + m_a(a_f - r_{tt}) - c_1 v_r - c_2 |v_r| v_r, \tag{6}$$

$$v_r = r_t - v_f, \qquad c_1, c_2 \geq 0. \tag{7}$$

Here $m_f = \rho_f A_f$ and m_a are displaced-fluid and added mass per unit length; a_f is the prescribed fluid acceleration. This is a reduced closure requiring calibration, not a solved Navier–Stokes interaction. In still fluid its drag power is $-c_1|r_t|^2 - c_2|r_t|^3$. In moving fluid it is dissipative relative to the fluid velocity and can do positive work on the rod. A constant added mass may be moved to the inertial side exactly once.

For a stationary wall, define penetration $\delta(r)$ positive inside the wall and $n_c = \nabla\delta$. One nonadhesive penalty model is

$$V_c = \frac{k_c}{m+1}[\delta]_+^{m+1}, \tag{8}$$

$$f_c = -k_c[\delta]_+^m n_c - c_c \mathbf{1}_{\{\delta>0\}}[\dot{\delta}]_+ n_c. \tag{9}$$

The elastic term is included through V_c and the second term dissipates power during compression. Friction can be added as a dissipative port. This switching law is used for force

accounting outside the contact-free KAM domain, not as an analytic perturbation through an impact.

Choose a configuration map $(r(s,q), D(s,q))$ that satisfies the essential boundary conditions and the rod constraints. A linear centreline ansatz alone is insufficient to specify rotational inertia. Define

$$r_i = \partial_{q_i} r, \qquad a_i = \mathrm{vee}(D^\mathsf{T} \partial_{q_i} D). \tag{10}$$

Then $\omega_b = \sum_i a_i \dot{q}_i$ and the reduced kinetic energy is

$$T_N = \tfrac{1}{2} \dot{q}^\mathsf{T} M(q) \dot{q}, \tag{11}$$

$$M_{ij}(q) = \int_0^L \left(\rho A r_i \cdot r_j + a_i^\mathsf{T} \mathcal{J}_b a_j \right) ds. \tag{12}$$

An added-fluid-mass contribution may be included when its consistent kinetic metric and associated inertial terms are specified. Otherwise hydrodynamic inertia remains in the generalized force balance and is not also included in M. Galerkin projection of the virtual-work identity gives

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + \nabla V_N(q) = Q_{\mathrm{nc}}, \tag{13}$$

$$Q_i = \int_0^L (f \cdot r_i + \tau \cdot D a_i) ds + Q_i^\partial. \tag{14}$$

Here $C\dot{q}$ denotes the Christoffel contribution of M, and V_N contains elastic, static magnetic and retained gravitational potentials. Microgravity reduces the last contribution; it does not eliminate fluid drag or establish a particular blood-flow profile. Relative-flow drag, moving-base work and contact losses are retained in Q_{nc}. A stationary-fluid quadratic drag, for example, has nonpositive power; a moving fluid may inject energy. Contact switching is excluded from the analytic neighbourhood used for KAM analysis.

For $M(q) \succ 0$, the Legendre transform yields

$$p = M(q)\dot{q}, \qquad H_N(q,p) = \tfrac{1}{2} p^\mathsf{T} M(q)^{-1} p + V_N(q). \tag{15}$$

In particular, $\dot{p}_i = -\frac{1}{2} p^\mathsf{T} \partial_{q_i}(M^{-1}) p - \partial_{q_i} V_N + (Q_{\mathrm{nc}})_i$. The configuration dependence of inertia is retained. Potential forces appear either in V_N or in Q_{nc}, never both.

2.2 Complex energy–dissipation representation with an explicit decoder

Energy and Rayleigh dissipation have different physical units. Let E_0 and t_0 be reference energy and time, and choose diagonal position and momentum scales Q_0, P_0 with $Q_0^\mathsf{T} P_0 = E_0 t_0 I$. Set $q = Q_0 x$, $p = P_0 y$, $\tau = t/t_0$ and

$$h(x,y) = H_N(Q_0 x, P_0 y)/E_0, \tag{16}$$

$$d(x,v) = \frac{t_0}{E_0} \mathcal{D}_N(Q_0 x, Q_0 v/t_0). \tag{17}$$

Here $v = dx/d\tau$ and $\mathcal{D}_N$ is a Rayleigh dissipation potential. The two entries of $h + id$ are now dimensionless. A complex network provides coupled features and two physically decoded real readouts,

$$\mathcal{H}_\theta(x,y,v) = h_\theta(x,y) + i d_\theta(x,v), \tag{18}$$

$$h_\theta = h_{\mathrm{prior}} + h_{\mathrm{res},\theta}. \tag{19}$$

One admissible dissipative head is

$$d_\theta(x,v) = \tfrac{1}{2} v^\mathsf{T} D_\theta(x) v, \qquad D_\theta(x) = L_\theta(x) L_\theta(x)^\mathsf{T} \succeq 0. \tag{20}$$

More general heads may be used if $d_\theta(x,\cdot)$ is differentiable, convex and minimized at zero. The decoder is

$$v = \partial_y h_\theta, \qquad x' = v, \tag{21}$$

$$y' = -\partial_x h_\theta - \partial_v d_\theta(x,v) + B(x)u. \tag{22}$$

The derivative of d_θ is taken with respect to its independent velocity argument before substituting $v = \partial_y h_\theta$. It is not a total derivative through this substitution.

Proposition 1. Power-consistent complex representation. *For a differentiable h_θ, the decoded system satisfies*

$$\frac{dh_\theta}{d\tau} = -v^\mathsf{T} \partial_v d_\theta(x,v) + u^\mathsf{T} B(x)^\mathsf{T} v + \partial_\tau h_\theta. \tag{23}$$

For the quadratic head, the first term equals $-v^\mathsf{T} D_\theta(x) v \le 0$. For the convex head it is also nonpositive.

Proof. Apply the chain rule to h_θ and substitute (22). The two canonical terms cancel. Convexity and the minimum at zero imply $v^\mathsf{T} \partial_v d_\theta(x,v) \ge d_\theta(x,v) - d_\theta(x,0) \ge 0$.

With $z = (x + iy)/\sqrt{2}$, the canonical convention is $\{z_j, \bar{z}_k\} = -i\delta_{jk}$ and

$$z' = -i\partial_{\bar{z}} h_\theta - \frac{i}{\sqrt{2}} \partial_v d_\theta + \frac{i}{\sqrt{2}} B(x) u. \tag{24}$$

This is a unified complex representation of conservative and nonconservative dynamics with a specified physical decoder. The rule $z' = -i\partial_{\bar{z}}(h_\theta + i d_\theta)$ would generally produce a different vector field and has no automatic dissipativity guarantee. Real analyticity in (x,y), and a controlled analytic extension to a complex neighbourhood, are the relevant requirements for analytic KAM. Holomorphicity

in z alone is not required of a nonconstant real energy $h_\theta(z,\bar z)$. In particular, ReLU or modReLU features require a separate smooth approximation or a theorem of appropriate regularity; their ordinary piecewise-smooth implementation is not analytic.

Hereafter, dimensionless energy, coordinates and time are denoted by H, (q,p) and t again; physical units are restored only through the stated scales.

3 Complex Hamiltonian learning

3.1 A fixed canonical chart

For a positive definite constant modal mass M and stiffness K, select U such that $U^{\mathsf T}MU = I$ and $U^{\mathsf T}KU = \Omega^2$, where $\Omega = \operatorname{diag}(\omega_1,\dots,\omega_n)$. The transformation

$$q = U\Omega^{-1/2}Q, \qquad p = MU\Omega^{1/2}P \tag{25}$$

is symplectic: its configuration and momentum blocks satisfy $S_q^{\mathsf T}S_p = I$. Consequently $H_2 = \frac{1}{2}\sum_j \omega_j(Q_j^2+P_j^2)$. The same chart is used for training, derivative errors and normal forms. Arbitrary independent standardization of q and p is not silently treated as canonical; optional training weights change the loss metric only.

Write $x = (Q,P)$ and use one complex normalization throughout:

$$z_j = \frac{Q_j + iP_j}{\sqrt{2}}, \quad I_j = z_j\bar z_j, \qquad \{z_j,\bar z_k\} = -i\delta_{jk}. \tag{26}$$

Wirtinger calculus gives $\partial_{\bar z_j} = (\partial_{Q_j} + i\partial_{P_j})/\sqrt{2}$ and therefore $\dot z = -i\partial_{\bar z}H$ for real H. This is a change of representation of a real symplectic system. Holomorphicity in z alone would exclude most real energy functions, including $H_2 = \sum_j \omega_j z_j\bar z_j$. The analyticity required by analytic KAM instead means a holomorphic extension of the real variables (Q,P) to a complex neighbourhood, in which z and $\bar z$ are independent linear combinations.

3.2 Architecture and objective

A shared complex feature map $F_\theta(z,\bar z,b)$, with magnetic configuration b, drives real readouts for H_θ and a dissipation factor L_θ. The latter defines $\mathcal{D}_\theta(q,v) = \frac{1}{2}\left\|L_\theta(q)^{\mathsf T}v\right\|^2$. The physical decoder in Section 2 determines the vector field. An affine complex layer has the real block realization

$$(A+iB)(a+ib) = (Aa - Bb) + i(Ba + Ab). \tag{27}$$

Thus complex weights impose a specific coupling, rather than furnishing an extra physical state. Smooth split activations $\sigma(a+ib) = \tanh(a) + i\tanh(b)$ are real analytic on bounded real domains and extend to a sufficiently small pole-free complex neighbourhood when weights are bounded. Their use does not justify assuming an arbitrarily wide analytic strip.

For observed dimensionless states and velocities, a suitable objective is

$$\widehat{\mathcal{L}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{\dot x}_i - f_\theta(x_i,u_i)\right\|^p + \lambda_E\widehat{\mathcal{L}}_E + \lambda_R\mathcal{R}(\theta). \tag{28}$$

Energy observations, if available, fix a gauge through $\widehat{\mathcal{L}}_E$; vector-field training alone identifies H only up to a constant. To infer conservative gradient error from the full decoder loss, errors in dissipation and inputs must first be bounded or measured and included in the label-error budget. They cannot be absorbed into a Hamiltonian residual merely because their magnitudes are small.

The reproducible experiments specialize this architecture to finite polynomial complex features. Products of complex affine forms and their conjugates generate all real quadratic and quartic monomials. They can be implemented by a finite network with polynomial activation, using $uv = ((u+v)^2 - (u-v)^2)/4$. A real readout produces a real polynomial energy; conjugate features enforce its reality. The prescribed monomial degrees exclude a spurious cubic term and, for the gray-box model, preserve the known quadratic part exactly. Fitting the output coefficients from energy derivatives is linear least squares. This is a controlled finite-feature CHNN, not a claim about the optimization of a deep unrestricted network.

3.3 What complex approximation theory establishes

Theorem 1 in the complex-reaction literature (Zhang et al. 2022) gives universal approximation for a specified reaction activation and an L^2 distributional criterion. Its comparison with real networks is tied to the stated function classes and resources. Such a theorem supplies neither higher derivative accuracy nor a universal advantage for magnetic catheter energies. In particular, piecewise reaction activations cannot be inserted unchanged into a smooth KAM proof.

There is nevertheless a direct smooth approximation route. A split-complex network with real-part output contains real smooth networks: for an input $(Q+iP)/\sqrt{2}$, choose first-layer complex weights $\sqrt{2}(A_Q - iA_P)$ to obtain real preactivation $A_QQ + A_PP + b$. Subsequent real weights

keep the real channel independent of the imaginary channel. Finite-order derivative approximation for smooth real networks (Hornik 1991) is thereby inherited as an existence result. Alternatively, on a compact box where a target has a convergent analytic power series, truncation gives a polynomial approximation with uniformly controlled derivatives. Those polynomials are realizable by the complex polynomial features used here. Training and a quantitative certificate are still required to establish that a particular fitted network attains the error needed for KAM.

4 From nonzero training loss to KAM persistence

4.1 A statistical bridge in the norm required by KAM theory

All Hamiltonians below are real-valued scalar functions, including when their parameters and hidden representations are complex. The realification of a complex parameter vector has dimension P. Let $\mathcal{D} = \mathbb{T}^n \times B$, where B is a bounded ball in action space, and write $d = 2n$. Lebesgue measure on $\mathcal{D}$ is denoted by dx. The following result completes the derivative-norm interface needed to combine the learning argument of Chen et al. (2022) with a finite-differentiability KAM theorem. It does not infer derivative smallness from function values alone.

Theorem 2. Uniform learning-to-C^ℓ bound. *Fix $d < p < \infty$, an integer $\ell \geq 1$, and an integer $s > \ell + d/p$. Let $X_1, \dots, X_N$ be independent samples with density $f_\mu \geq m_0 > 0$ on $\mathcal{D}$. Let $H_\star$ be the target Hamiltonian and $\{H_\vartheta : \vartheta \in \Theta\}$ the hypothesis class, with $\Theta \subset [-B_\Theta, B_\Theta]^P$. Assume that for all $x \in \mathcal{D}$ and $\vartheta, \eta \in \Theta$,*

$$\|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p \leq Q, \tag{29}$$

$$\|\nabla H_\vartheta(x) - \nabla H_\eta(x)\|_p \leq L_g \|\vartheta - \eta\|_\infty. \tag{30}$$

After fixing the additive gauge by

$$e_\vartheta = H_\vartheta - H_\star - \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} (H_\vartheta - H_\star)\, dx,$$

$$\sup_{\vartheta \in \Theta} \|e_\vartheta\|_{W^{s,p}(\mathcal{D})} \leq M_s, \tag{31}$$

suppose that M_s is a valid uniform bound. Define

$$\widehat{R}_N(\vartheta) = \frac{1}{N}\sum_{i=1}^N \|\nabla H_\vartheta(X_i) - \nabla H_\star(X_i)\|_p^p, \tag{32}$$

$$L_{\mathrm{par}} = pQ^{p-1}L_g, \qquad L_{\max} = Q^p. \tag{33}$$

For any fixed $0 < a \leq 2B_\Theta$ and $0 < \delta < 1$, put

$$\Gamma_N(a,\delta) = 2L_{\mathrm{par}}a + \frac{L_{\max}}{\sqrt{2N}}\sqrt{P\log(1 + 2B_\Theta/a) + \log(1/\delta)}, \tag{34}$$

$$S_N(\vartheta) = \widehat{R}_N(\vartheta) + \Gamma_N(a,\delta), \tag{35}$$

$$G_N(\vartheta) = C_{\mathrm{P}}\, m_0^{-1/p} S_N(\vartheta)^{1/p}, \tag{36}$$

$$\alpha = \frac{\ell - 1 + d/p}{s - 1} \in (0,1). \tag{37}$$

Here C_{P} is a Poincare constant chosen so that a mean-zero function satisfies $\|e\|_{W^{1,p}} \leq C_{\mathrm{P}}\|\nabla e\|_{L^p}$. With probability at least $1 - \delta$, simultaneously for every $\vartheta \in \Theta$,

$$\|e_\vartheta\|_{C^\ell(\overline{\mathcal{D}})} \leq E_N(\vartheta) := C_{\mathrm{GN}}\left[G_N(\vartheta) + M_s^\alpha G_N(\vartheta)^{1-\alpha}\right]. \tag{38}$$

C_{GN} depends only on the domain and (d, p, s, ℓ). We use $\|e\|_{C^\ell} = \max_{|a| \leq \ell} \sup |D^a e|$; equivalent conventions only change the displayed constants. The event is uniform over the hypothesis class and therefore includes parameters selected using these same training data.

Proof. The loss $r_\vartheta(x) = \|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p^p$ lies in $[0, L_{\max}]$ and is L_{par}-Lipschitz in the ℓ^∞ parameter metric. Partitioning the parameter cube into cells of side length a and choosing a representative from each nonempty cell gives an a-net with at most $(1 + 2B_\Theta/a)^P$ points. The one-sided Hoeffding inequality and a union bound control $R(\eta) - \widehat{R}_N(\eta)$ at every net point by the square-root term in Γ_N, where $R(\eta) = \int r_\eta\, d\mu$. The population and empirical losses each change by at most $L_{\mathrm{par}}a$ between a parameter and its net representative. Consequently $R(\vartheta) \leq S_N(\vartheta)$ uniformly.

The density lower bound gives $\|\nabla e_\vartheta\|_{L^p(dx)} \leq m_0^{-1/p} S_N(\vartheta)^{1/p}$. The gauge in (31) and Poincare's inequality then yield $\|e_\vartheta\|_{W^{1,p}} \leq G_N(\vartheta)$. Apply the Gagliardo–Nirenberg inequality to $g = \nabla e_\vartheta$, with derivative orders $j = \ell - 1$, $m = s - 1$, and both endpoint Lebesgue exponents equal to p. Its interpolation identity is

$$0 = \frac{j}{d} + \alpha\left(\frac{1}{p} - \frac{m}{d}\right) + \frac{1-\alpha}{p},$$

which gives the stated α. The strict inequality $s > \ell + d/p$ ensures $\alpha < 1$ and avoids the critical Sobolev endpoint. Extension from the smooth bounded action domain, together with periodic charts in the angular variables, gives

$$\|e\|_{C^\ell} \leq C_{\mathrm{GN}}\left(\|e\|_{W^{1,p}} + \|e\|_{W^{1,p}}^{1-\alpha}\|e\|_{W^{s,p}}^{\alpha}\right).$$

Lower-order derivatives are included by the inhomogeneous term and the $W^{1,p} \hookrightarrow C^0$ embedding, which uses $p > d$. Substitution proves (38).

For instance, one may select an integer $\ell > 2(\tau + 1)$, $p > 2n$, and $s = \ell + 1$. This makes every exponent explicit, although the resulting interpolation bound can be conservative. Bounds on M_s, Q, L_g follow from bounded parameters and derivatives of smooth activations on a compact domain; the corresponding bounds for $H_\star$ must also be justified. Merely declaring the functions smooth does not provide these quantitative bounds. Alternatively, a loss containing all spatial derivatives of the vector field through order ℓ controls $W^{\ell+1,p}$ directly, and Sobolev embedding then gives a C^ℓ error with exponent $1/p$, without interpolation.

Corollary 3. KAM transfer for the target and the trained Hamiltonian. *Let $n \geq 2$, $\tau > n - 1$ and an integer $\ell > 2(\tau + 1)$ be fixed. Suppose that, in one specified canonical action–angle chart,*

$$H_\star = h_0(I) + P_\star(I, \varphi), \qquad H_\vartheta = h_0(I) + P_\vartheta(I, \varphi).$$

Assume $h_0 \in C^{\ell+2}(\overline{B})$ with finite norm, and $\nabla h_0 : B \to \Omega$ is a diffeomorphism with bounded inverse in C^ℓ. Fix $\gamma > 0$ and set

$$\Omega_{\gamma,\tau} = \left\{ \omega \in \Omega : \begin{array}{l} |k \cdot \omega| \geq \gamma |k|_1^{-\tau} \ (0 \neq k \in \mathbb{Z}^n), \\ \operatorname{dist}(\omega, \partial\Omega) \geq \gamma \end{array} \right\}. \tag{39}$$

Let $c_{\mathrm{K}} > 0$ be the constant in Theorem A of Bounemoura (2020) for this domain and integrable Hamiltonian. If a certified bound $\beta_\vartheta \geq \|P_\vartheta - \overline{P}_\vartheta\|_{C^\ell}$ satisfies

$$\beta_\vartheta + E_N(\vartheta) < c_{\mathrm{K}} \gamma^2, \tag{40}$$

then, on the event in Theorem 2, both $H_\star$ and H_ϑ possess invariant Lagrangian tori for every frequency in $\Omega_{\gamma,\tau}$. The flow on each torus is C^1-conjugate to the linear flow with that frequency. For a piecewise smooth frequency-domain boundary, the complement of each family has measure at most $C\gamma$. In particular, each family has positive measure when γ is small enough.

Proof. Write $\overline{P}$ for the spatial mean over $\mathcal{D}$. The gauge-fixed difference satisfies

$$P_\star - \overline{P}_\star = P_\vartheta - \overline{P}_\vartheta - e_\vartheta.$$

Thus the two perturbation norms are at most β_ϑ and $\beta_\vartheta + E_N(\vartheta)$, respectively. Constants added to a Hamiltonian do not change its vector field. Apply the cited finite-smoothness KAM theorem separately to the two Hamiltonians with the same h_0, domain, and Diophantine set. The theorem supplies tori of class $C^{\tau+1}$ with Lipschitz dependence as invariant graphs on frequency. The two systems need not have identical invariant embeddings.

The certified budget includes the physical perturbation from h_0, any normal-form remainder, and the learning error. A small learning error alone does not establish that the physical system is close to an integrable one. Theorem A in the published 2020 version of Bounemoura (2020) gives the measure bound $C\gamma$; the $C\gamma^2$ measure bound printed in the 2018 preprint should not be used.

4.2 A coefficient certificate for a complex polynomial CHNN

The polynomial readout used in the experiments yields a stronger certificate that avoids a dimension-dependent interpolation loss. Let $\psi_1, \ldots, \psi_m$ be real-valued basis functions on the real phase domain. Complex features may be converted into this basis by taking their real and imaginary parts and collecting equal monomials. The certificate applies to the effective coefficients of the actual trained Hamiltonian, irrespective of how they are parameterized inside the network.

Proposition 4. Deterministic global certificate from sampled gradients. *Suppose*

$$H_\vartheta = H_{\mathrm{prior}} + \sum_{j=1}^{m} c_{\vartheta,j} \psi_j,$$
$$H_\star = H_{\mathrm{prior}} + \sum_{j=1}^{m} c_{\star,j} \psi_j + R.$$

The basis contains no constant gauge direction. Let

$$G_N = \frac{1}{N} \sum_{i=1}^{N} D\psi(X_i)^\top D\psi(X_i),$$
$$\lambda_N = \lambda_{\min}(G_N) > 0,$$

where the jth column of $D\psi$ is $\nabla\psi_j$. For gradient observations $Y_i = \nabla H_\star(X_i) + \xi_i$, define

$$r_N = \left(\frac{1}{N} \sum_i \|\nabla H_\vartheta(X_i) - Y_i\|_2^2 \right)^{1/2},$$
$$t_N = \left(\frac{1}{N} \sum_i \|\nabla R(X_i) + \xi_i\|_2^2 \right)^{1/2}.$$

Then

$$\|c_\vartheta - c_\star\|_2 \le \frac{r_N + t_N}{\sqrt{\lambda_N}}. \tag{41}$$

For any function norm $\|\cdot\|_{\mathcal{X}}$ satisfying the triangle inequality, put $K_{\mathcal{X}} = (\sum_j \|\psi_j\|_{\mathcal{X}}^2)^{1/2}$. If $\|R\|_{\mathcal{X}} \le \varepsilon_R$, then

$$\|H_\vartheta - H_\star\|_{\mathcal{X}} \le K_{\mathcal{X}} \frac{r_N + t_N}{\sqrt{\lambda_N}} + \varepsilon_R. \tag{42}$$

The norm may be C^ℓ on a real compact domain or the supremum norm on a prescribed complex neighborhood where every basis function and R is holomorphic. No independent sampling assumption is needed.

Proof. Let $\Delta c = c_\vartheta - c_\star$. The empirical norm of $D\psi\Delta c$ is $(\Delta c^\top G_N \Delta c)^{1/2}$. Since $D\psi\Delta c = (\nabla H_\vartheta - Y) + (\nabla R + \xi)$ at every sample, the triangle inequality gives $\sqrt{\lambda_N}\|\Delta c\|_2 \le r_N + t_N$. Finally, $\|\sum_j \Delta c_j \psi_j - R\|_{\mathcal{X}} \le K_{\mathcal{X}}\|\Delta c\|_2 + \varepsilon_R$ by Cauchy–Schwarz.

For raw monomials $\psi_\nu(x) = x^\nu$, an explicit C^ℓ bound on $[-r, r]^d$ is obtained from

$$\|\psi_\nu\|_{C^\ell} \le \max_{|a|\le\ell,\, a\le\nu} \frac{\nu!}{(\nu-a)!}\, r^{|\nu|-|a|}.$$

The analytic sup norm on $\{|x_j| \le r\} \subset \mathbb{C}^d$ is $r^{|\nu|}$. For all 35 degree-four monomials in four variables, $K_{\mathrm{an}} \le \sqrt{35}\, r^4$. If a canonical chart $x = \Phi(I, \varphi)$ maps an action–angle complex domain into this polydisc, the same analytic bound holds after composition. For a C^ℓ certificate after composition, the derivative bounds of Φ must instead be included by the chain rule. Unmodeled physical terms require the tail bounds t_N and ε_R; setting them to zero is justified in the exact polynomial benchmark only.

The statistical result requires independent state samples with the stated coverage density. Consecutive states on a trajectory do not satisfy this assumption automatically. Accordingly, the trajectory-based experiments below use the deterministic Gram certificate, which requires no sampling independence. Their implementation divides both squared residuals and the Gram matrix by $4N$ rather than N; the coefficient bound is unchanged provided the noise norm uses that same scalar-observation normalization.

5 A mechanically derived benchmark and its nonlinear twist

5.1 A transparent Galerkin specialization

To make every coefficient and every datum recoverable, consider two transverse polarizations $w(s) = \phi(s)(q_1, q_2)$ of an axially restrained extensible beam. In the small-slope stretching model the longitudinal strain is $u_s + \frac{1}{2}|w_s|^2$. Minimizing its axial energy with $u(L) - u(0) = 0$ gives a constant strain and

$$V_{\mathrm{stretch}} = \frac{EA}{8L}\left(\int_0^L |w_s|^2\, \mathrm{d}s\right)^2. \tag{43}$$

Indeed the axial resultant is constant, and integration of u_s determines that constant. After modal mass and time normalization, linear bending and a locally conservative static magnetic stiffness give the fully specified benchmark

$$H_*(q,p) = \tfrac{1}{2}(p_1^2 + p_2^2 + q_1^2 + 2q_2^2) + \tfrac{\beta}{4}(q_1^2 + q_2^2)^2, \tag{44}$$

$$\beta = 0.15, \qquad (\omega_1, \omega_2) = (1, \sqrt{2}). \tag{45}$$

Before nondimensionalization, write the physical quartic coefficient as $K_4 = (EA/(8L))(\int \phi_s^2\, \mathrm{d}s)^2$ in mass-normalized coordinates. The dimensionless coefficient β is obtained after applying the declared coordinate, energy and time scales. The selected dimensionless values specify a test model, not identified material constants. The restraint in (43) is essential: an axially free cantilever does not generally have this stretching energy. A real catheter with a different support, vessel constraint or large rotation requires the more general reduction in Section 2. Static magnetic stiffness is included once in the potential; variable magnetic loading remains an explicit input.

5.2 Local invariant tori for the trained quartic Hamiltonian

Proposition 5. Exact quartic normalization and a sixth-order remainder. *Let $x = (Q, P) \in \mathbb{R}^{2n}$ and*

$$H(x) = H_2(x) + H_4(x), \quad H_2 = \frac{1}{2}\sum_{j=1}^n \omega_j(Q_j^2 + P_j^2),$$

where H_4 is any real homogeneous quartic polynomial. Suppose $k \cdot \omega \neq 0$ for $0 < |k|_1 \le 4$. Set $w_j = (Q_j + iP_j)/\sqrt{2}$ and $I_j = w_j\bar{w}_j$. Writing $H_4 = \sum_{|a|+|b|=4} h_{ab} w^a \bar{w}^b$, define

$$Z_4(I) = \sum_{a=b} h_{aa} I^a, \qquad \chi_4 = -\sum_{a\neq b} \frac{h_{ab}}{i(a-b)\cdot\omega} w^a \bar{w}^b.$$

The time-one Hamiltonian flow Φ of χ_4 is real and canonical near the origin and satisfies the exact identity

$$H \circ \Phi = \omega \cdot I + Z_4(I) + R_6(Q,P), \qquad R_6 = O(\|x\|^6). \tag{46}$$

In particular, the transformed system retains the full remainder.

Proof. For the Poisson convention $\{F,G\} = \partial_Q F \cdot \partial_P G - \partial_P F \cdot \partial_Q G$, one has $\{H_2, w^a \bar{w}^b\} = i(a-b)\cdot\omega\, w^a \bar{w}^b$. Thus $\{H_2, \chi_4\} = Z_4 - H_4$. Conjugate symmetry of the coefficients makes χ_4 real on real phase space. Let Φ^t denote its flow. Taylor's formula along the flow gives

$$R_6 = \int_0^1 \big[t\{H_4,\chi_4\} + (1-t)\{Z_4,\chi_4\}\big] \circ \Phi^t \, dt. \quad (47)$$

Both brackets have degree six, and $\Phi^t(x) = x + O(\|x\|^3)$. This proves the order and the exact identity without truncating the flow. For an explicit local analytic bound, let C_χ, C_4, C_Z be coefficient ℓ^1 norms of the three quartic polynomials in the Cartesian variables. If $32C_\chi r^2 < 1$, the flow starting on the complex polydisc of radius r remains on the polydisc of radius $2r$ for $0 \le t \le 1$: each component of X_{χ_4} there is bounded by $32C_\chi r^3$. The polynomial bracket bound $\|\{F_m, G_k\}\|_{\mathrm{coeff},1} \le mk\|F_m\|_{\mathrm{coeff},1}\|G_k\|_{\mathrm{coeff},1}$ now yields

$$\|\Phi - \mathrm{id}\|_r \le 32C_\chi r^3, \qquad \|R_6\|_r \le 512C_\chi(C_4 + C_Z)r^6.$$

An additional analytic term of order six or higher in the original Hamiltonian contributes its composition with Φ to the remainder.

Corollary 6. KAM tori of the actual trained polynomial CHNN. *Let $n \ge 2$. Assume that both the target and trained Hamiltonians have the same H_2 in Proposition 5, and that their learned corrections are homogeneous quartics. For $\nu \in \{\star, \vartheta\}$, write their quartic averages as $Z_{4,\nu}(I) = \frac{1}{2} I^\top A_\nu I$. If $A_\star$ and A_ϑ are positive definite, then both original Hamiltonian flows possess positive-measure families of invariant Lagrangian tori arbitrarily close to the origin. For each system, the relative measure of the union of its invariant tori in sufficiently small neighborhoods tends to one as the neighborhood radius tends to zero. This assertion concerns the complete target and trained quartic Hamiltonians, including their nonzero normal-form remainders.*

Proof. The preceding proposition verifies the finite-order normal-form hypothesis of the local KAM theorem near an elliptic equilibrium (Delshams and Gutiérrez 1996, Theorem 3). The theorem's isoenergetic nondegeneracy holds because

$$\det \begin{pmatrix} A_\nu & \omega \\ \omega^\top & 0 \end{pmatrix} = -\det(A_\nu)\,\omega^\top A_\nu^{-1}\omega \neq 0.$$

Its order-four nonresonance condition is exactly the hypothesis already verified above. Apply the theorem to each complete Hamiltonian and pull the tori back by its exact canonical transformation. Canonical transformations preserve phase volume. The local theorem's relative-measure estimate tends to zero for the excluded set. Moreover, $\Phi = \mathrm{id} + O(r^3)$ sandwiches an image of a radius-r ball between balls of radii $r \pm O(r^3)$. Their relative volume difference is $O(r^2)$, so the density-one conclusion also holds in the original coordinates. The positive-definiteness assumption can be replaced by the corresponding Kolmogorov or isoenergetic nondegeneracy conditions with the appropriate version of local KAM; positive definiteness provides a direct certificate for both conditions in the benchmark.

For the two-mode benchmark, canonical balancing is $Q_j = \sqrt{\omega_j} q_j$, $P_j = p_j/\sqrt{\omega_j}$. The target quartic is

$$H_{4,\star} = \frac{\beta}{4}\left(\frac{Q_1^2}{\omega_1} + \frac{Q_2^2}{\omega_2}\right)^2,$$
$$A_\star = \begin{pmatrix} \dfrac{3\beta}{4\omega_1^2} & \dfrac{\beta}{2\omega_1\omega_2} \\ \dfrac{\beta}{2\omega_1\omega_2} & \dfrac{3\beta}{4\omega_2^2} \end{pmatrix}.$$

For $\omega = (1, \sqrt{2})$ and $\beta = 0.15$, $\det A_\star = 0.003515625$ and $\lambda_{\min}(A_\star) \simeq 0.02434571$. Nonresonance is exact: for every nonzero integer vector k,

$$|k_1 + \sqrt{2}k_2| = \frac{|k_1^2 - 2k_2^2|}{|k_1 - \sqrt{2}k_2|} \ge \frac{1}{\sqrt{2}|k|_1}.$$

For raw quartic monomial coefficients c in the order (Q_1, Q_2, P_1, P_2), the full trained twist is obtained by the linear map

$$A_{11} = 3c_{Q_1^4} + c_{Q_1^2P_1^2} + 3c_{P_1^4}, \quad (48)$$
$$A_{22} = 3c_{Q_2^4} + c_{Q_2^2P_2^2} + 3c_{P_2^4}, \quad (49)$$
$$A_{12} = c_{Q_1^2Q_2^2} + c_{Q_1^2P_2^2} + c_{P_1^2Q_2^2} + c_{P_1^2P_2^2}. \quad (50)$$

All other quartic monomials have zero angular average, but remain in R_6 through normalization. Since $\|A_\vartheta - A_\star\|_{\mathrm{op}} \le \sqrt{19}\|c_\vartheta - c_\star\|_2$, Proposition 4 supplies the sufficient condition

$$\sqrt{19}\frac{r_N + t_N}{\sqrt{\lambda_N}} < \lambda_{\min}(A_\star)$$

for positive definiteness of the trained twist. A direct interval bound on $\lambda_{\min}(A_\vartheta)$ can be sharper.

The corollary certifies a nonempty sufficiently small amplitude regime. Appendix D strengthens this result with a fully explicit annular certificate for the target and the actual fitted models: for $0 < a \le 2^{-225}$, at least $949/1024$ of a precisely defined transformed annulus is occupied by invariant tori. The enclosing balanced-coordinate bound $|(Q,P)|_\infty < 6a$ is extremely conservative. The measure

denominator is that annulus, not the enclosing ball; the much larger simulation amplitudes are not certified by this bound.

6 Reproducible numerical experiments

6.1 Protocol and matched model classes

Learning observations and the following two-mode tests are generated from (45). There are three independent training replicates, with seeds 17, 43 and 91. Each uses ten trajectories, 33 states per trajectory and a training horizon $T = 16$. Initial actions are sampled uniformly from $[0.02, 0.22]^2$ and angles from $[0, 2\pi]^2$. Independent componentwise $\mathrm{Uniform}[-0.003, 0.003]$ noise is added to the exact vector-field labels. A separate set of eight test trajectories has 801 states per trajectory on $[0, 80]$. Initial states, clean labels, noisy labels, coefficients and predicted trajectories are all saved. DOP853 with relative and absolute tolerances 10^{-11} and 10^{-13} supplies the reference trajectories; these are numerical solutions of a declared model, not physical measurements.

The factorial comparison uses BR (real features, no quadratic prior), BC (complex features, no prior), GR (real features, fixed prior) and GC (complex features, fixed prior). Without the prior there are ten quadratic and 35 quartic real coefficients; with the prior only the 35 quartic coefficients are fitted. The complex polynomial basis is transformed explicitly into the same real monomial span. A ridge coefficient 10^{-10} penalizes the common effective real coefficients, and the output weights are solved by least squares. Thus equal performance between the paired representations is an expected control for implementation consistency. Changing to a different activation, depth or fixed parameter budget would be a different approximation comparison.

For each fitted model the field metric is the Euclidean norm of all test vector-field errors divided by that of the reference field. The trajectory metric is computed separately for each orbit,

$$e_i = \left(\frac{\sum_k \|x_{\theta,i}(t_k) - x_{*,i}(t_k)\|_2^2}{\sum_k \|x_{*,i}(t_k)\|_2^2} \right)^{1/2}, \qquad (51)$$

and then averaged over eight test trajectories. Energy excursion is the maximum relative change of the true Hamiltonian along each predicted orbit, also averaged over test orbits. The reported dispersion is the sample standard deviation across the three training replicates, not across correlated time samples.

Table 1. Held-out errors, mean ± sample SD over three training replicates. The two models grouped in each row agree to numerical precision. All percentages are relative errors, not torus-retention percentages.

Models	Field error (%)	Trajectory error (%)
BR, BC	0.0782 ± 0.0170	1.123 ± 0.543
GR, GC	0.0711 ± 0.0171	0.745 ± 0.293

Table 2. Twist of the actual trained GC energies at the main noise level. The determinant and eigenvalue use the full quartic coefficient vector.

Seed	$\det A_\theta$	$\lambda_{\min}(A_\theta)$
17	0.00322280	0.0225605
43	0.00384444	0.0264865
91	0.00314547	0.0217599
Target	0.003515625	0.0243457

The prior improves all three seeds, with only a small change for seed 17, and reduces the mean trajectory metric by 33.7% in this experiment (Table 1). With three replicates this is a descriptive comparison. The largest paired real–complex coefficient difference is 2.45×10^{-15}, and the complex-to-real feature-gradient discrepancy is 1.12×10^{-16}. The corresponding mean true-energy excursions are 9.56×10^{-4} for BR/BC and 7.09×10^{-4} for GR/GC. Figure 1 shows all replicates rather than only their mean.

6.2 The learned Hamiltonians satisfy local KAM hypotheses

The quartic non-action divisors comprise 32 monomials; their smallest absolute value is $\sqrt{2} - 1$. Exact irrationality establishes nonresonance at every integer order, whereas this finite enumeration checks the specific normal-form calculation. The fixed quadratic prior and zero cubic terms hold by construction for GC.

The exported decimal coefficients define reproducible mathematical polynomials. Exact rational evaluation of the learned twist's leading principal minors verifies positive definiteness for all three models. Together with Corollary 6, this establishes positive-measure invariant-torus families in a sufficiently small neighbourhood for the target and each of these actual trained GC models. It does not identify the full training domain as that neighbourhood.

The measured GC Gram eigenvalues are 5.31×10^{-4}, 9.84×10^{-4} and 1.22×10^{-3}. At the main noise level, the training-based coefficient bounds are 0.203, 0.151 and 0.134, respectively. They bound the true coefficient errors but are too conservative to establish twist positivity through the error bound alone. The direct algebraic check above and the following noise-budget experiment are distinct certificates.

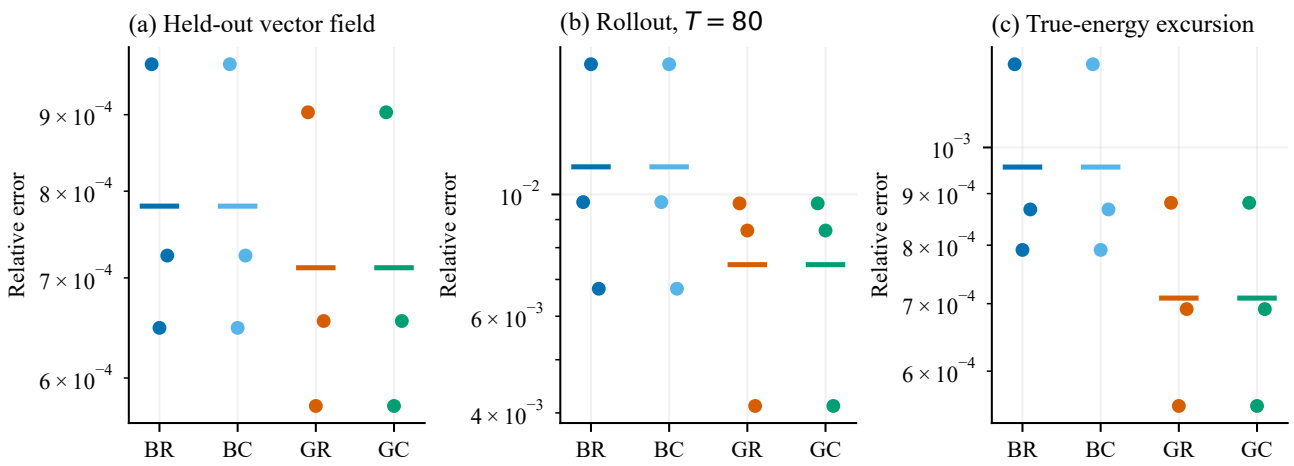


Figure 1. Identification with matched polynomial hypothesis classes. Points show the three training replicates and horizontal marks their means. BR/BC and GR/GC agree because the real and complex features span the same class and use the same effective coefficient penalty. The physical prior is the independently varied factor.

A nonzero-loss certificate that passes. We retain the main experiment above and repeat GC coefficient fitting on the same three training designs at noise bounds 10^{-4}, 5×10^{-5} and 10^{-5}. This is a declared noise-sensitivity study, not a replacement of the less favourable original noise level. Every elementary arithmetic operation used to enclose the Gram matrix is rounded outwards. For an approximate inverse R of the enclosed Gram matrix, the bound

$$\lambda_{\min}(G) \geq \frac{1 - \|I - RG\|_\infty}{\|R\|_\infty} > 0 \tag{52}$$

follows from the inverse-residual identity, positive semidefiniteness and symmetry of G. Interval residual evaluation also encloses the fitted loss. Clean-label evaluation and arithmetic budgets are included in the effective noise bound. These enclosures use IEEE-754 double precision with outward `nextafter` inflation; no underflow or overflow occurs in this calculation.

For the exact target twist, $\lambda_{\min}(A_*) \geq \det A_* / \operatorname{tr} A_* = 1/48$. Combining this lower bound with Proposition 4 and the twist coefficient map gives the sufficient test

$$\sqrt{19}\frac{\overline{r}_N + \overline{\eta}}{\sqrt{\underline{\lambda}_N}} < \frac{1}{48}. \tag{53}$$

Bars denote outward bounds, and the loss and Gram use the common $4N$ normalization. At noise bound 5×10^{-5} all three runs pass (Table 3), with strictly nonzero training loss. At 10^{-5} all pass; at 10^{-4} only seed 91 passes this particular sufficient test. Failure of the test is inconclusive about the actual twist or torus existence.

Thus this experiment closes the chain from a measured nonzero derivative loss, a declared noise budget and verified excitation to positive learned twist and the invariant tori of Corollary 6. The radius-0.8 complex polydisc used in the exported norm estimates is not asserted to satisfy a quantitative KAM smallness threshold.

Table 3. Verified loss-to-twist budget at noise bound 5×10^{-5}. Every entry in the last column is below the target lower bound $1/48 = 0.0208333\ldots$. This certifies the local KAM hypotheses, not a fixed experimental amplitude or a global safe region.

Seed	$\underline{\lambda}_N$	Training MSE	Twist-error bound
17	$4.6560\,10^{-4}$	$7.8458\,10^{-10}$	0.015759
43	$8.0926\,10^{-4}$	$8.2519\,10^{-10}$	0.012063
91	$1.0668\,10^{-3}$	$7.8268\,10^{-10}$	0.010406

6.3 Integration of the same learned Hamiltonian

We integrate the seed-17 GC energy with RK4, implicit midpoint, a fourth-order central generating-function scheme (Central4) and two-stage fourth-order Gauss–Legendre integration. All four methods use exactly the same exported learned energy, initial condition and time units. Central4 and its order and symplecticity are derived in Appendix B. Implicit iterations stop when the infinity norm of consecutive updates falls below 2×10^{-13}; failure to converge within 80 iterations raises an error. Observed update residuals are retained in the data record.

Steps $h = 0.24, 0.12, 0.06, 0.03$ over $T = 24$ give approximately fourth-order state convergence for RK4, Central4 and Gauss4, and second-order convergence for midpoint (Figure 3). Over $T = 1200$ at $h = 0.12$, the maximum relative learned-energy excursions are 2.05×10^{-3}, 2.53×10^{-5}, 1.28×10^{-7} and 3.10×10^{-8}, respectively. Gauss4 performs best among these methods for this orbit and step.

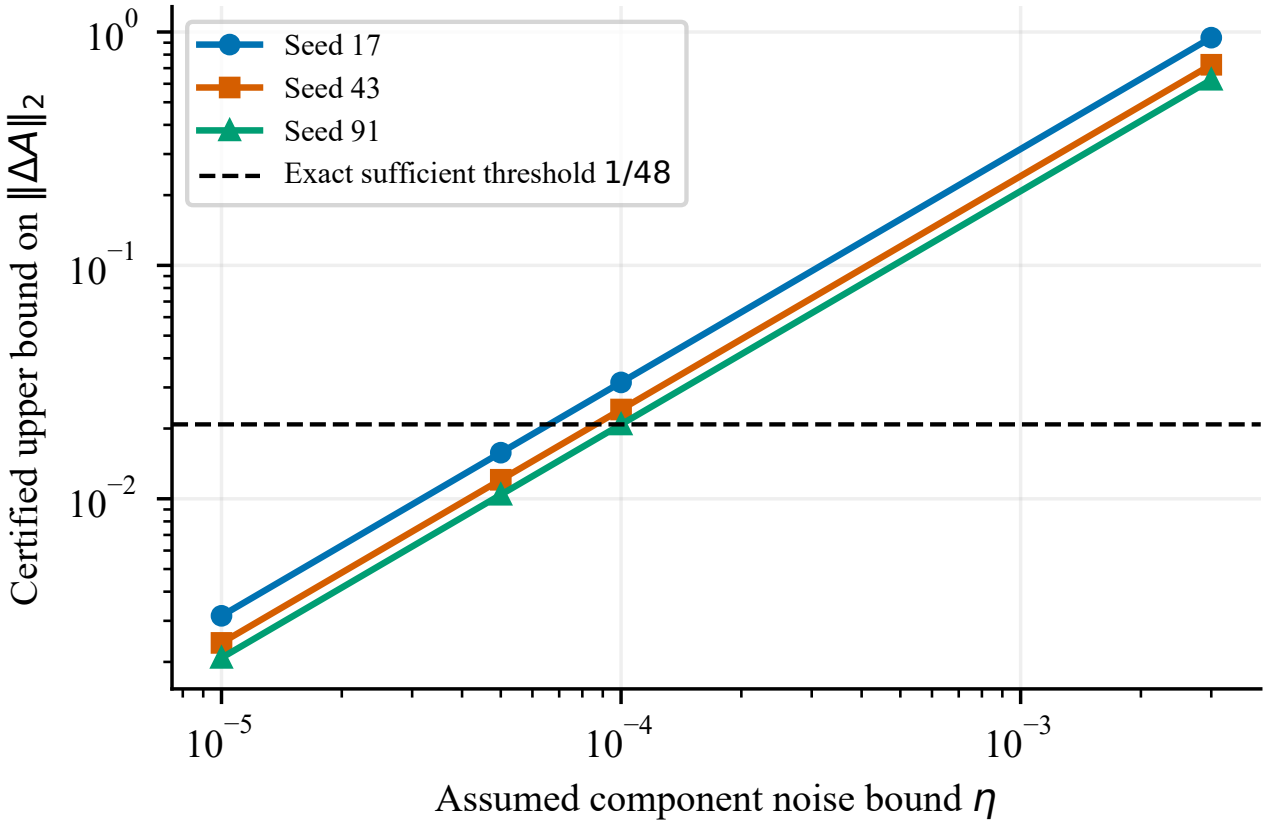


Figure 2. A verified sufficient loss-to-twist budget under decreasing derivative-label noise. The main noisy experiment is retained. Crossing below the stated twist-eigenvalue lower bound establishes the local KAM hypotheses through the deterministic training certificate; crossing above it is inconclusive.

The generating-function method is therefore reported as a verified structural construction, without an unsupported superiority claim.

6.4 Resonance diagnostics and dissipative operation

To examine sensitivity to spectral spacing and nonlinear strength, we vary ω_2/ω_1 across 27 values in $[1, 1.65]$ and $\beta \in \{0.03, 0.15, 0.45\}$, for 81 configurations. Gauss4 uses $h = 0.04$ and $T = 600$. The action excursion and change of phase-fit frequencies between the two half-windows are saved. Sections at $Q_2 = 0, P_2 > 0$ use interpolated crossings and are shown for several initial conditions (Figure 4). The sections are trajectory diagnostics and are not claimed to lie on a common energy surface.

The exactly isotropic case $\omega_1 = \omega_2$ has a radial potential and conserved angular momentum. It is an integrable resonant example, so a large exchange of the coordinate actions near 1:1 cannot be labelled chaos from that observation. The scan illustrates sensitivity and the need for nonresonance hypotheses; it does not estimate an exact KAM-breakdown threshold or a percentage of surviving tori.

The port experiment restores $\mathcal{D} = 0.04 \|\dot{q}\|^2/2$ and, in the forced case, a generalized input with each component bounded by $a = 0.012$. From the same benchmark energy,

$$H(t) \le H(0) + \frac{2a^2}{4(0.04)} t = 0.184510875 + 0.0018t. \tag{54}$$

Because $V(q) \ge \|q\|^2/2$, this also bounds displacement by $\|q(t)\| \le \sqrt{2H(0) + 0.0036t}$. At $T = 160$ the bound is 0.97213. Therefore a declared safe reduced-coordinate ball of radius one contains the trajectory throughout this horizon for the stated initial energy and input bounds. This is a dimensionless geometric constraint; a vessel-specific clearance uses the reconstruction in Section 7. The corresponding sufficient componentwise input limit is

$$a < \left[\frac{2(0.04)(1/2 - H(0))}{160}\right]^{1/2} \simeq 0.01256. \tag{55}$$

The integrated power-balance residual is below 4.4×10^{-12} in both runs, and no saved state exceeds the analytic envelope. Figure 5 compares the actual trajectory with that bound. The dissipation and input coefficients here

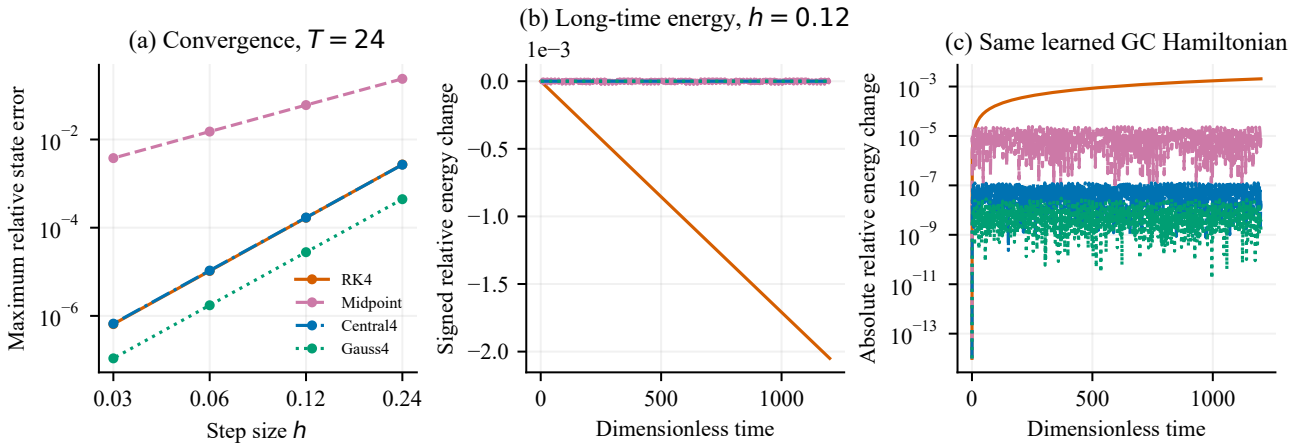


Figure 3. Integrator comparison on the same learned GC Hamiltonian. (a) Step-size refinement against a high-accuracy reference. (b,c) Signed and absolute relative energy changes over 1200 dimensionless time units. Small energy excursion does not by itself prove torus persistence or small phase error.

are specified, rather than estimated from the conservative training data.

6.5 Identification of the dissipative readout

We additionally fit the imaginary dissipation readout, after fixing either the known conservative target H_* or the actual seed-17 GC energy. This sequential experiment uses new data and does not infer damping from the earlier conservative trajectories. The true dimensionless damping is $D_* = \mathrm{diag}(0.04, 0.06)$. Six unforced damped trajectories on $[0, 16]$, each with 81 states, form the training set. Initial actions are sampled from $[0.03, 0.18]^2$ and angles from $[0, 2\pi]^2$ using a saved seed. Independent componentwise $\mathrm{Uniform}[-3 \times 10^{-4}, 3 \times 10^{-4}]$ noise is added to balanced-coordinate vector-field labels.

In the balanced chart, the actual decoded physical-coordinate velocity is $v_j = \partial_{P_j} H / \sqrt{\omega_j}$. The dissipative decoder is

$$\dot{P}_j = -\partial_{Q_j} H + \frac{-d_j v_j + u_j}{\sqrt{\omega_j}}. \tag{56}$$

For the fitted GC energy, v_j must be computed from its full momentum derivative; substituting the target-only expression $\sqrt{\omega_j} P_j$ would discard learned momentum-dependent terms. For the unforced training states, set $r_{ij} = \sqrt{\omega_j}(\widehat{P}_{ij} + \partial_{Q_j} H(x_i))$. Nonnegative least squares has the explicit solution

$$\widehat{d}_j = \max\left(0, -\frac{\sum_i v_{ij} r_{ij}}{\sum_i v_{ij}^2}\right), \qquad \mathcal{D}_\theta(v) = \tfrac{1}{2} \sum_j \widehat{d}_j v_j^2. \tag{57}$$

Thus $L_\theta = \mathrm{diag}(\sqrt{\widehat{d}_j})$ provides a globally positive-semidefinite dissipation head. With a known conservative energy, the bounded label noise gives the deterministic estimate

$$|\widehat{d}_j - d_{*,j}| \le \frac{\delta \sqrt{\omega_j} \sum_i |v_{ij}|}{\sum_i v_{ij}^2}, \qquad \delta = 3 \times 10^{-4}. \tag{58}$$

Errors in a learned conservative readout contribute an additional bias; the noise-only bound is not applied to that case.

Four independently initialized forced trajectories on $[0, 60]$, with 601 saved states each, are held out. Their two input components have amplitude 0.012, frequencies 0.7 and 1.1, and distinct saved phases. Field errors are evaluated on the reference states; the trajectory metric is the mean relative L^2 state error over the four trajectories. The results of this single seeded identification experiment are The relative damping errors are 0.175% and 0.0857% with H_*, and 0.116% and 0.157% with GC-17. For both fitted decoders the dissipative power $-\sum_j \widehat{d}_j v_j^2$ is nonpositive by construction. Independently integrating total port power gives a maximum energy-balance residual below 1.04×10^{-11} over the forced tests. External input can still increase total energy. This experiment completes both readouts of the complex energy–dissipation model in a specified finite hypothesis class; it is not a demonstration of joint deep-network optimization or a KAM theorem for the forced damped trajectories.

6.6 Independent clamped-free magnetic-beam verification

To test the mechanical discretization separately from the axially constrained two-mode benchmark, we implemented a clamped-free Euler–Bernoulli beam with cubic Hermite elements and consistent translational mass. This is a newly

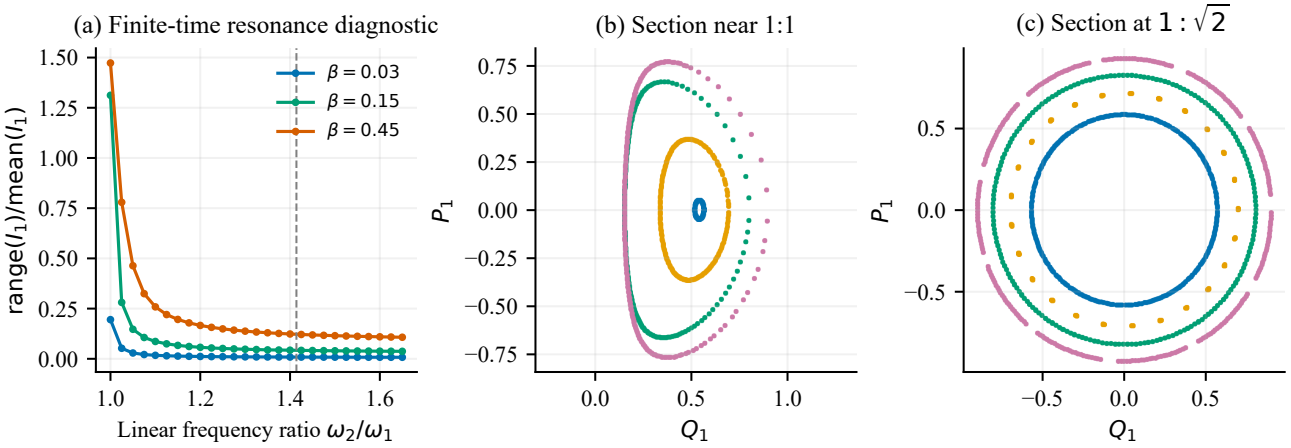


Figure 4. Finite-time spectral and phase-space diagnostics for the declared benchmark. (a) Relative excursion of a coordinate action versus linear frequency ratio. (b,c) Poincaré sections from separate trajectories. The 1:1 case is rotationally integrable; these plots must not be interpreted as a binary map of chaos or surgical safety.

run numerical verification, not COMSOL or experimental data. The illustrative, uncalibrated SI parameters are $L = 0.08$ m, circular radius $r = 0.0006$ m, $E = 5 \times 10^6$ Pa and $\rho = 1100$ kg m^{-3}. The base displacement and slope are fixed. A tip dipole $\mu = 2 \times 10^{-5}$ A m^2 interacts with the uniform field

$$B_0 = 0.03 \text{ T}, \tag{59}$$

$$B_y(t) = 0.006[1 - \exp(-(t/0.25)^2)] \sin(\Omega t) \text{ T}, \tag{60}$$

$$\Omega = 9.444949 \text{ s}^{-1}. \tag{61}$$

We set the field gradient and gravity to zero. At fixed tip axial position $x = L$, the small-slope approximation identifies the tip angle with $\theta = w_s(L)$. The magnetic potential and torque are

$$V_{\text{mag}} = -\mu(B_0 \cos\theta + B_y \sin\theta), \tag{62}$$

$$\tau_{\text{mag}} = \mu(B_y \cos\theta - B_0 \sin\theta). \tag{63}$$

There is no translational magnetic force in this uniform-field case. With distributed stationary-medium Rayleigh drag, the semi-discrete equation is

$$M\ddot{q} + \alpha M\dot{q} + Kq = e_\theta \tau_{\text{mag}}, \qquad \alpha = 0.555585 \text{ s}^{-1}. \tag{64}$$

Rotary inertia, axial extension, contact and moving-fluid forcing are omitted in this independent small-slope check.

The first four undriven elastic frequencies converge against the analytic cantilever roots as the mesh increases from 8 to 16 to 32 elements. Their relative errors at 32 elements are 7.92×10^{-9}, 3.21×10^{-7}, 2.51×10^{-6} and 9.62×10^{-6}. A static tip-moment check reproduces $w(L) = \tau L^2/(2EI)$ to a maximum relative error of 4.91×10^{-12}. Over the 8 s driven experiment, the 8- and 16-element tip responses differ from the 32-element response by at most 6.72×10^{-8} m and 4.05×10^{-9} m, respectively.

On the 32-element mesh, retaining 2, 4 and 8 mass-normalized modes gives maximum tip-displacement errors of 1.115%, 0.687% and 0.377% relative to the full 64-mode response peak. Corresponding tip-angle errors are 5.656%, 2.843% and 1.420%, showing that the localized magnetic torque requires a stronger modal-resolution check than tip displacement alone. The full response reaches a tip-displacement peak of 3.0903 mm and a sampled maximum slope of 0.05736 rad.

The time-dependent energy satisfies the independently checked identity

$$H(t) - H(0) = \underbrace{\int_0^t -\mu \dot{B}_y \sin\theta \, ds}_{W_{\text{in}}} - \underbrace{\int_0^t \alpha \dot{q}^{\mathsf{T}} M \dot{q} \, ds}_{W_{\text{d}}}. \tag{65}$$

The maximum balance residual divided by the maximum energy activity is 5.83×10^{-8}. Tightening solver tolerances by a factor of 100 reduces this residual to 9.64×10^{-10} and changes the tip trajectory by at most 3.44×10^{-10} m. All mesh and modal errors are empirical convergence diagnostics; they do not supply a rigorous continuum reconstruction bound ε_N. No KAM conclusion is asserted for this driven, damped experiment.

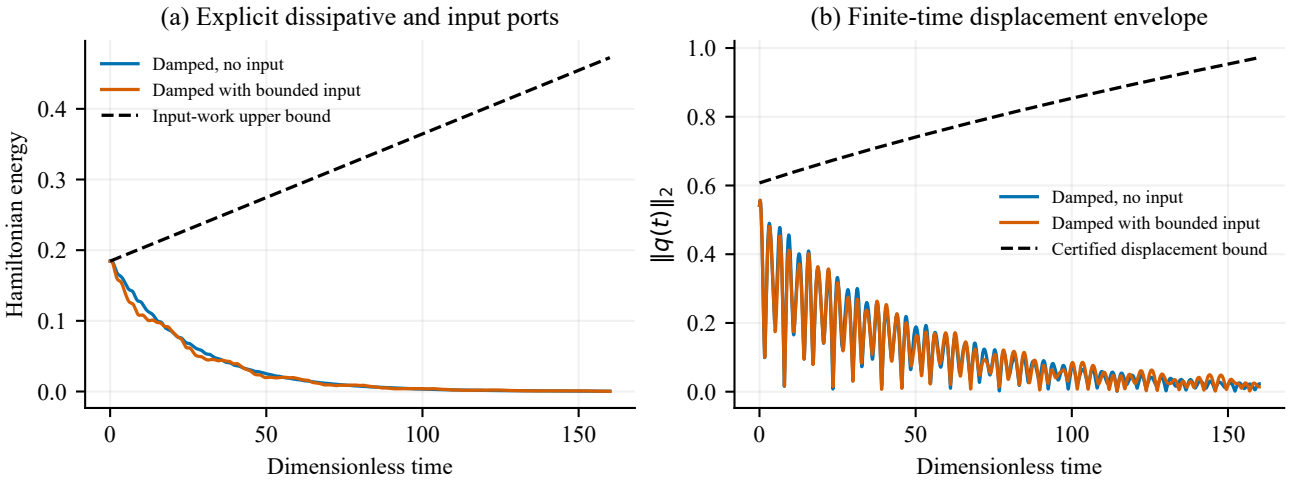


Figure 5. Dissipation, bounded input and a finite-time operating envelope. The upper bounds follow from the energy inequality for the model and include all times in the declared horizon. The saved-state comparisons check numerical consistency with that analytic statement.

7 From invariant motion to operational constraints

7.1 From a certified conservative torus to physical clearance

Let $f_0 = J\nabla H_{\mathrm{ref}}$ be the conservative reduced vector field used in the normal-form and KAM calculation. It must be the conservative part of the same reduced model and canonical chart used above. Suppose the KAM hypotheses, including nondegenerate twist, nonresonance and the specified remainder norm, have established an invariant embedding $K : \mathbb{T}^n \to \mathbb{R}^{2n}$ satisfying

$$f_0(K(\varphi)) = DK(\varphi)\omega. \tag{66}$$

If a computed embedding $\widehat{K}$ has a validated error bound $\|K - \widehat{K}\|_\infty \le \varepsilon_K$, its geometric clearance can be bounded by that of $\widehat{K}$ minus the corresponding reconstruction error. An unvalidated residual is a numerical diagnostic and does not by itself establish ε_K.

The physical reduced motion is represented on a neighbourhood U of this torus by

$$Z' = f_0(Z) + r(Z,\tau), \qquad \sup_{U\times[0,T]} \|r\| \le \eta. \tag{67}$$

The budget η includes learned-field error, open-port forces, model-reduction error and external disturbances; it is not identified with the training loss. Let f_0 be L_f-Lipschitz on U. For initial error $e_0 = \|Z(0) - K(\varphi_0)\|$, define

$$E(\tau) = \begin{cases} e_0 e^{L_f\tau} + \eta(e^{L_f\tau} - 1)/L_f, & L_f > 0, \\ e_0 + \eta\tau, & L_f = 0. \end{cases} \tag{68}$$

Let $r_N(s,x)$ reconstruct the centreline and assume

$$\sup_s \|r_N(s,x) - r_N(s,\widetilde{x})\| \le L_r\|x - \widetilde{x}\|, \tag{69}$$

$$\sup_s \|r_{\mathrm{physical}}(s,\tau) - r_N(s,x(\tau))\| \le \varepsilon_N. \tag{70}$$

The second estimate must be supplied by a reduction/validation bound; it is an additional requirement for a conclusion about the continuum rod, beyond a statement about its reduced model.

Proposition 7. Finite-time clearance certificate. *Suppose the reconstructed reference torus has centreline clearance, after subtracting catheter radius, at least $c_* > 0$ from a stationary vessel wall. If*

$$E(T) < \mathrm{dist}(K(\mathbb{T}^n), \partial U), \qquad L_r E(T) + \varepsilon_N < c_*, \tag{71}$$

then the trajectory remains in U and avoids wall contact for $0 \le \tau \le T$. Its clearance is bounded below by $c_* - L_r E(\tau) - \varepsilon_N$.

Proof. Subtract the equation for $K(\varphi_0 + \omega\tau)$ from the equation for Z. Until a possible first exit from U, the upper right derivative of their distance is at most $L_f e + \eta$. Gronwall's inequality gives $e(\tau) \le E(\tau)$. The first strict inequality prevents that exit by continuity. Distance to a closed stationary wall is a 1-Lipschitz function of position. The reconstruction and reduction bounds therefore imply the stated clearance estimate, and the second inequality excludes contact.

This certificate converts a verified torus into a finite-time operation constraint for the full forced, damped model.

Table 4. Sequential identification of the dissipation readout. Both models use the same independent damped training data and forced test set.

Conservative readout	$\widehat{d}_1$	$\widehat{d}_2$	Field error (%)	Trajectory error (%)
Known H_*	0.0400700	0.0599486	0.00397	0.0769
Learned GC-17	0.0400465	0.0600944	0.0449	0.3828

For $L_f > 0$, a prescribed allowance $a = (c_* - \varepsilon_N)/L_r > e_0$ gives the explicit budget

$$\eta < \frac{L_f(a - e_0 e^{L_f T})}{e^{L_f T} - 1}, \tag{72}$$

provided the right side is positive and the domain condition also holds. Bounds on B, damping and field modulation can then convert η to admissible input magnitudes. This is a sufficient operation bound, not an exact chaos threshold.

7.2 A complementary energy-barrier certificate

A stronger conclusion is available when the same mechanical model has a useful energy barrier at the unsafe boundary. Let $\mathcal{C}$ be the connected safe configuration component, including a geometric margin for catheter radius and validated reduction error, and let

$$h(x, y) = \tfrac{1}{2} y^{\mathsf{T}} M(x)^{-1} y + v_0(x), \qquad E_{\mathrm{b}} = \inf_{x \in \partial \mathcal{C}} v_0(x). \tag{73}$$

Assume $M \succ 0$ and a smooth solution exists up to a potential first boundary crossing. Let $P_{\mathrm{net}}(t)$ denote the full right-hand side of the power identity, including explicit time dependence. Require a nonnegative bound on every prefix of the energy injection,

$$W_T \geq \sup_{0 \leq t \leq T} \int_0^t P_{\mathrm{net}}(s)\, \mathrm{d}s. \tag{74}$$

A bound only on the final net work at T would not exclude an earlier excursion. If this prefix bound satisfies

$$h(Z(0)) + W_T < E_{\mathrm{b}}, \tag{75}$$

then no unsafe boundary is reached by time T. Indeed, at a first crossing nonnegative kinetic energy would imply $h \geq E_{\mathrm{b}}$, contradicting the integrated power identity. The infimum is over every part of the unsafe boundary, not selected sampled wall points. If it lies below the initial energy, this certificate is inconclusive and the torus-tube criterion remains available.

For $D(x) \succeq d_* I$ with $d_* > 0$, a generalized forcing error $\|w\| \leq \overline{w}$ gives

$$h' = -v^{\mathsf{T}} D v + v^{\mathsf{T}} w \leq \frac{\overline{w}^2}{4 d_*}, \qquad W_T = \frac{T \overline{w}^2}{4 d_*}. \tag{76}$$

Any explicit time-dependent energy contribution is added to W_T. If a learned natural Hamiltonian $\widehat{h}$ has the same nonnegative kinetic structure and a uniform energy error δ_h against the physical model, with its potential barrier computed on the same configuration boundary, a conservative transferable condition is

$$\widehat{h}(Z(0)) + 2\delta_h + W_T < \widehat{E}_{\mathrm{b}}. \tag{77}$$

For a general neural energy without a natural kinetic/potential split, one must instead bound its infimum over the full admissible phase-space boundary. Neither a finite loss nor an energy plot provides this global lower bound.

KAM persistence establishes quasiperiodicity on the invariant tori covered by its theorem. It does not establish that every initial state is nonchaotic, that a fixed percentage such as 95% is safe, or that torus breakdown equals tissue injury. The clearance and energy inequalities provide the additional geometric and physical connection needed for safety claims, while retaining explicit conditions that can be tested for the stated model.

8 Discussion

The framework connects CHNNs and KAM theory through testable hypotheses. The complex output has an explicit conservative–dissipative decoder. The learned Hamiltonian is compared with its physical target in one canonical chart. Its derivative loss is promoted to an appropriate global norm, or its finite-feature design matrix supplies a deterministic alternative. The quartic twist is computed for the learned network itself. This supports a theorem about the learned conservative system, with nonzero training error, rather than a numerical analogy to a separate polynomial.

The two learning routes address different uses. The smooth-network statistical result accommodates richer hypothesis classes and quantified truncation errors, but its high derivative constants can make its sufficient error

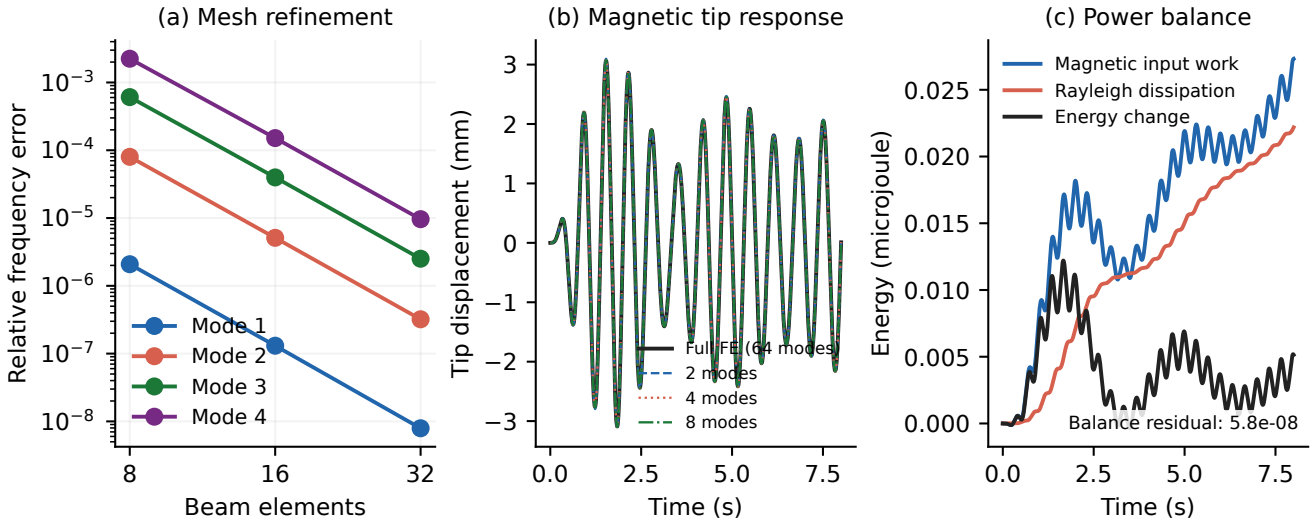


Figure 6. Independent clamped-free beam verification: (a) modal-frequency convergence; (b) driven magnetic tip response under modal truncation; (c) magnetic input work, Rayleigh dissipation and stored-energy change. The SI parameters are declared illustrative values.

threshold conservative. The finite-feature result is readily computable and requires persistent excitation, encoded by the smallest Gram eigenvalue. Model misspecification must be included as a tail; a very small fitted residual cannot reveal an unobserved term outside the hypothesis class. The fully specified polynomial benchmark makes this tail exactly zero for that experiment. It does not set the tail to zero for a physical catheter.

The local KAM conclusion is stronger than finite-time visual regularity and narrower than global stability: a positive-measure family of quasiperiodic invariant tori persists at sufficiently small amplitudes. It does not assert that every initial condition is on one of those tori, that resonance implies chaos, or that a numerical frequency scan tests all integer resonance orders. In more than two degrees of freedom, invariant tori do not generally partition an energy surface. These distinctions are necessary when translating a dynamical statement into an operating constraint. The clearance and energy results supply that translation under their own stated force and geometry bounds.

Microgravity removes a gravitational load term; it does not by itself establish incompressible axisymmetric flow, suppress contact, or produce twist. Static magnetic fields can modify equilibrium and stiffness, while varying fields inject power. Quantitative limits in tesla, millimetres or millimetres per second require the physical scales, magnetic map, wall geometry and reduction error. A measured magnetic-field change also requires either the frozen-field assumptions of the conservative analysis or the time-dependent power term of the safety analysis. We therefore distinguish dimensionless identification experiments from an illustrative SI beam-discretization check and supply explicit formulas for subsequent calibration.

Several extensions are concrete. The rod discretization should be refined until a certified or independently bounded truncation error can enter the transfer and safety inequalities. A catheter-specific magnetic map must establish the available generalized-force directions and input saturation. Fluid forces and contact laws require validation in the relevant geometry; discontinuous impacts lie outside the smooth domain used for classical KAM. A time-periodic or quasiperiodic operating protocol calls for an appropriate forced or dissipative KAM theorem with its additional hypotheses, rather than treating its nonconservative force as an arbitrary Hamiltonian perturbation.

9 Conclusion

A mechanically grounded CHNN can support KAM stability analysis when its representation, training loss and perturbation norm are linked explicitly. We provide that link through smooth statistical estimates and a deterministic finite-feature certificate, and compute the nonlinear twist of the actual learned conservative model. A complete analytic remainder estimate and explicit KAM constants also yield a conservative annular amplitude and measure certificate. Reproducible two-mode experiments demonstrate the resulting identification and verification workflow. Dissipative power and clearance estimates complement invariant-torus persistence with sufficient operational constraints. This establishes a coherent route from rod mechanics to learned quasiperiodic structure and bounded operating motion; physical calibration remains

necessary to instantiate that route for a specific magnetic catheter in microgravity.

Data and code availability

The accompanying reproducibility directory contains the complete dimensionless model, data-generation and fitting code, saved observations and coefficients, evaluation records and figure sources. Every reported numerical result is generated by these scripts from the declared mathematical models.

Declarations

Author identities, affiliations, funding, competing interests and author contributions must be completed by the submitting authors. This article reports mathematical analysis and computer-generated mechanical data; no human, animal or in-orbit experiment was performed.

A Integrator structure and reproducibility checks

For a real smooth Hamiltonian, implicit midpoint is

$$x_{k+1} = x_k + hJ\nabla H\big((x_k + x_{k+1})/2\big). \tag{78}$$

Let $A = JH''(\bar{x})$. Differentiation gives $D\Phi_h = (I - hA/2)^{-1}(I + hA/2)$. Since $A^{\mathsf{T}}J + JA = 0$, this Cayley transform satisfies $D\Phi_h^{\mathsf{T}} J D\Phi_h = J$ whenever the implicit equation defines a differentiable local map. The proof assumes the nonlinear step is solved exactly; residual tolerances add a numerical defect. Explicit RK4 is not symplectic in general. For the separable benchmark, the kick–drift–kick Störmer–Verlet map is also symplectic, because each substep is the exact flow of the kinetic or potential Hamiltonian. A general learned momentum-dependent quartic no longer admits that separable split, so the same formula cannot be reused without justification.

Energy error and symplecticity are distinct diagnostics. Finite-time energy plots are evaluated against the specified reference Hamiltonian and accompanied by a step-size check. A small integration error does not certify a Diophantine inequality, nor does a positive finite-time Lyapunov estimate alone establish asymptotic chaos. These facts determine the labels used in the numerical figures.

B The fourth-order central generating-function scheme

Let $f = J\nabla H$, $A = H''$ and define a central generating function

$$S_h(x) = hH(x) + \frac{h^3}{24} f(x)^{\mathsf{T}} A(x) f(x). \tag{79}$$

The map is specified implicitly by

$$x_+ - x_- = J\nabla S_h\big((x_- + x_+)/2\big). \tag{80}$$

Canonical maps generated in central variables are a standard geometric integration construction (Feng et al. 1989; Hairer et al. 2006). Here every derivative is computed from the same learned scalar Hamiltonian. The identity

$$J\nabla(f^{\mathsf{T}} A f) = f''[f,f] - 2(f')^2 f \tag{81}$$

follows by differentiating both occurrences of f, using $f' = JA$ and $J^{\mathsf{T}} = -J$. It gives the displacement implemented in the reproducibility code:

$$g_h = hf + \frac{h^3}{24}\big(f''[f,f] - 2(f')^2 f\big). \tag{82}$$

The Hessian S_h'' is symmetric, so the midpoint Cayley-transform argument in Appendix A proves exact symplecticity of the locally defined map. Since $S_{-h} = -S_h$, interchanging its endpoints proves reversibility.

To verify order, expand an exact orbit about its temporal midpoint x_0. Its endpoint midpoint m and endpoint difference satisfy

$$m = x_0 + \tfrac{1}{8}h^2 f'(x_0) f(x_0) + O(h^4), \tag{83}$$

$$x_+ - x_- = hf(x_0) + \tfrac{1}{24}h^3\big(f''[f,f] + (f')^2 f\big)(x_0) + O(h^5). \tag{84}$$

Substituting $x_0 = m - \frac{1}{8}h^2 f'(m) f(m) + O(h^4)$ gives $x_+ - x_- = g_h(m) + O(h^5)$, proving fourth order under the required smoothness and local solvability assumptions. The comparison includes Gauss4 because equal formal order and exact symplecticity do not imply equal errors or computational cost.

C Complex-gradient descent with a strict descent margin

Theorem 3 of Zhang et al. (2022) motivates complex-gradient optimization, but a stationary-point conclusion

requires a strict descent margin. The following convention-explicit statement also applies to a smooth CHNN regression objective.

Proposition 8. Complex descent. *Let $\mathcal{L} : \mathbb{C}^P \simeq \mathbb{R}^{2P} \to \mathbb{R}$ be bounded below and have an L-Lipschitz real gradient on every segment traversed by the iterates. For $w = a + ib$, define $G(w) = \nabla_a \mathcal{L} + i\nabla_b \mathcal{L} = 2\partial\mathcal{L}/\partial\bar{w}$. Choose a constant $\rho \in \mathbb{C}$ with $\kappa = \operatorname{Re}\rho - (L/2)|\rho|^2 > 0$, and update $w_{t+1} = w_t - \rho G(w_t)$. Then*

$$\mathcal{L}(w_{t+1}) \le \mathcal{L}(w_t) - \kappa\|G(w_t)\|_2^2, \qquad \lim_{t\to\infty} \|G(w_t)\|_2 = 0,$$

and

$$\min_{0\le t<T} \|G(w_t)\|_2^2 \le \frac{\mathcal{L}(w_0) - \inf \mathcal{L}}{\kappa T}.$$

Proof. Under realification, multiplication by $\rho = u + iv$ is $uI + vJ_c$, with $J_c^\top = -J_c$ and $J_c^\top J_c = I$. Hence the update has inner product $-u\|G\|^2$ with the real gradient and squared norm $|\rho|^2\|G\|^2$. The real descent lemma gives the first inequality. Summing it and using the lower bound for $\mathcal{L}$ gives $\kappa\sum_{t\ge0}\|G(w_t)\|^2 \le \mathcal{L}(w_0) - \inf\mathcal{L}$. Square summability proves convergence of the gradient norm and the minimum bound. The result asserts convergence to stationarity in gradient norm, not convergence to a global minimum or to a particular parameter vector.

For a real step this condition is $0 < \rho < 2/L$ under the stated convention. The case $\rho = 0$, allowed by a non-strict printed step-size interval, cannot imply gradient convergence. Nonsmooth complex activations also need separate treatment; a smooth polynomial CHNN satisfies the differentiability requirements, with a Lipschitz gradient bound on any bounded iterate region.

D A fully explicit amplitude certificate

The local result can be made fully quantitative for the target and the trained GC Hamiltonians. The following conservative certificate uses the explicit analytic KAM theorem of Chierchia and Koudjinan (2021, Theorems 1 and 3, Remark A.4). Here matrix norms are induced by the vector maximum norm, whereas $\lambda_{\min}$ denotes a Euclidean eigenvalue.

Proposition 9. An explicit annular KAM certificate. *Let*

$$H = \frac{1}{2}\sum_{j=1}^{2}\omega_j(Q_j^2 + P_j^2) + H_4(Q,P),$$
$$H_4 = \sum_{|\mu|=4} c_\mu (Q,P)^\mu,$$

where the coefficients are real and

$$\min_{0<|k|_1\le4} |k\cdot\omega| \ge \frac{2}{5}, \qquad \sum_{|\mu|=4}|c_\mu| \le 1. \tag{85}$$

The minimum is over integer vectors $k \in \mathbb{Z}^2$. Let the quartic angular average be $Z_4(I) = \frac{1}{2}I^\top AI$, and assume

$$\begin{aligned} A \succ 0, \qquad \|A\|_\infty \le 1, \\ \|A^{-1}\|_\infty \le 128, \qquad \lambda_{\min}(A) \ge \frac{1}{128}. \end{aligned} \tag{86}$$

For every $0 < a \le 2^{-225}$ define

$$I^0 = a^2(1,1), \qquad r = \frac{a^2}{2},$$
$$D_a = \{I \in \mathbb{R}^2 : |I - I^0|_\infty < r\}.$$

Let Φ be the exact quartic normalizing transformation, with $H \circ \Phi = \omega\cdot I + \frac{1}{2}I^\top AI + R_6$, and set

$$\Psi(I,\theta) = (Q_1,Q_2,P_1,P_2),$$
$$Q_j = \sqrt{2I_j}\cos\theta_j, \qquad P_j = -\sqrt{2I_j}\sin\theta_j,$$
$$\Omega_a = \Phi\big(\Psi(D_a\times\mathbb{T}^2)\big).$$

The original Hamiltonian H has a union $\mathcal{T}_a$ of invariant Lagrangian two-tori in Ω_a such that

$$\begin{aligned} \frac{\operatorname{meas}(\mathcal{T}_a)}{\operatorname{meas}(\Omega_a)} \ge \frac{949}{1024}, \\ \Omega_a \subset \{(Q,P) : |(Q,P)|_\infty < 6a\}. \end{aligned} \tag{87}$$

The denominator in this statement is the explicitly defined transformed annulus Ω_a, whose volume is $(2\pi)^2a^4$.

Proof. We give every domain and smallness bound used in the application. Write $\|F\|_{\mathrm{coeff},1}$ for a Cartesian polynomial coefficient norm. A homogeneous degree-four polynomial acquires at most a factor four in coefficient norm under either of the linear substitutions $w_j = (Q_j + iP_j)/\sqrt{2}$ and its inverse. Choose $\{H_2,\chi_4\} = Z_4 - H_4$ and $\Phi = \phi^1_{\chi_4}$, with $\{F,G\} = F_Q\cdot G_P - F_P\cdot G_Q$. The homological equation and (85) therefore give

$$C_\chi := \|\chi_4\|_{\mathrm{coeff},1} \le 40,$$
$$C_Z := \|Z_4\|_{\mathrm{coeff},1} \le 16, \qquad C_H := \|H_4\|_{\mathrm{coeff},1} \le 1.$$

The coefficient bracket estimate $\|\{F_j,G_k\}\|_{\mathrm{coeff},1} \le jk\|F_j\|_{\mathrm{coeff},1}\|G_k\|_{\mathrm{coeff},1}$, applied to the exact remainder identity

$$R_6 = \int_0^1 \big[t\{H_4,\chi_4\} + (1-t)\{Z_4,\chi_4\}\big]\circ\Phi^t\,dt,$$

implies

$$\begin{aligned} \|R_6\|_b \le 512 C_\chi (C_H + C_Z) b^6 < 2^{19} b^6, \\ \Phi^t(\mathbb{D}_b^4) \subset \mathbb{D}_{2b}^4, \end{aligned} \tag{88}$$

provided $32 C_\chi b^2 < 1$. Here $\mathbb{D}_b^4$ is the complex Cartesian polydisc. The flow inclusion follows by integration of $|X_{\chi_4}|_\infty \le 32 C_\chi b^3$ on $\mathbb{D}_{2b}^4$. At $b = 3a$ the required inequality is $11520 a^2 < 1$, which follows from the stated bound on a.

Use the complex action polydisc $|I - I^0|_\infty < r$ and angle strip $|\operatorname{Im}\theta|_\infty < s = 1/2$. Every action lies in the disc centered at a^2 with radius $a^2/2$; thus this domain avoids zero and admits a single holomorphic square-root branch, positive on its real interval. Moreover,

$$|\Psi(I,\theta)|_\infty \le \sqrt{3}\, a e^{1/2} < 3a.$$

Consequently the complete transformed Hamiltonian is analytic there and can be written as $K(I) + P(I,\theta)$ with

$$K = \omega \cdot I + \tfrac{1}{2} I^\top A I, \qquad \mathcal{P} := \|P\|_{r,1/2} < 2^{29} a^6.$$

No term of the normal-form remainder has been discarded.

Apply the cited KAM theorem with its perturbation multiplier equal to one, dimension $d = 2$, Diophantine exponent $\tau = 2$, $\nu = \tau + 1 = 3$, analytic widths $s = 1/2$ and $s_* = 1/4$, and singleton center set $\{I^0\}$. In its notation choose

$$\begin{aligned} R = r = \frac{a^2}{2}, \qquad \rho = \frac{r}{128}, \\ \alpha = \frac{r}{2^{17}} = \frac{a^2}{2^{18}}, \qquad \vartheta = ML \le 2^7, \\ M = \|A\|_\infty \le 1, \qquad L = \|A^{-1}\|_\infty \le 128. \end{aligned}$$

For completeness, the explicit constant c_* in that theorem obeys

$$c_* < 2^{280} \quad \text{when } d = 2,\ \tau = 2. \tag{89}$$

This bound follows directly from the formulas in its Remark A.4; the details are recorded below. Its constant c_0 is greater than one. The two required smallness inequalities hold because

$$\frac{L\alpha}{\rho} \le 2^{-3} < c_0, \tag{90}$$

$$\frac{M\mathcal{P}}{\alpha^2} < 2^{65} a^2 \le 2^{-385} < 2^{-384} \le \frac{(s - s_*)^{31}}{c_* \vartheta^6}. \tag{91}$$

Thus the theorem supplies an invariant torus set in $D_a \times \mathbb{T}^2$ and a frequency reparametrization with Lipschitz deviation at most $1/(4d) = 1/8$.

We evaluate its measure estimate explicitly. Put $m = 1/128$. For each $k \in \mathbb{Z}^2 \setminus \{0\}$, the non-Diophantine strip

$$\{I \in D_a : |k \cdot (\omega + AI)| < \alpha / |k|_1^2\}$$

has area at most $8r\alpha/(m|k|_1^3)$: its normal vector is Ak, $|Ak|_2 \ge m|k|_2 \ge m|k|_1/\sqrt{2}$, and every parallel section of the square D_a has length at most $2\sqrt{2}r$. There are $4j$ integer vectors with $|k|_1 = j$, so

$$\operatorname{meas}(\mathcal{R}_\alpha) \le \frac{32 r\alpha}{m} \sum_{j=1}^{\infty} j^{-2} \le \frac{64 r\alpha}{m}.$$

The boundary strip has area $4(r+\rho)^2 - 4(r-\rho)^2 = 16 r\rho$. Since $\operatorname{meas}(D_a) = 4r^2$, the relative excluded measure in Theorem 3 of the cited reference is at most

$$\left(1 + \frac{1}{4}\right)^2 \left(\frac{4\rho}{r} + \frac{16\alpha}{mr}\right) = \frac{25}{16}\left(\frac{1}{32} + \frac{1}{64}\right) = \frac{75}{1024}.$$

Both Ψ on this real annulus and the exact normalizing map Φ preserve Liouville volume and conjugate the corresponding Hamiltonian flows. The negative sign in the displayed expression for P_j gives $\dot\theta_j = \partial_{I_j} K$ with the convention $\dot Q = H_P$, $\dot P = -H_Q$. Mapping the torus set through $\Phi \circ \Psi$ proves the measure claim for the original Hamiltonian. Finally, $\Psi(D_a \times \mathbb{T}^2) \subset \mathbb{D}_{3a}^4 \cap \mathbb{R}^4$ and (88) imply $\Omega_a \subset \{|(Q,P)|_\infty < 6a\}$.

Explicit constant audit. To distinguish the reference's constants from polynomial norms, denote the constants C_j in its Remark A.4 by C_j. For $d = 2$ one has exactly

$$\int_{\mathbb{R}^2} |y|_1^k e^{-|y|_1}\, dy = 4(k+1)!.$$

Substitution of $\tau = 2$, $\nu = 3$, followed by $e^{-3} < 1$ and $\sqrt{2} < 2$, gives the following rational upper bounds:

j	0	1	2	3	4	5	6
$\mathsf{C}_j \le$	2^{22}	2^{11}	2^{7}	2^{24}	2^{25}	2^{8}	2^{2}

j	7	8	9	10	11	12
$\mathsf{C}_j \le$	2^{55}	2^{44}	2^{102}	2^{14}	2^{30}	2^{31}

The upper bounds for $\mathsf{C}_0, \mathsf{C}_1, \mathsf{C}_3, \mathsf{C}_4, \mathsf{C}_7, \mathsf{C}_8, \mathsf{C}_9$ are strict. In the reference, c_* is the maximum of

$$\begin{aligned} & 2^{49} 27 e^{-3} \mathsf{C}_5^2 \mathsf{C}_6 \mathsf{C}_7 \mathsf{C}_8, \\ & \left(2^{-1/2} 214 \cdot 27 e^{-3} \mathsf{C}_4 \mathsf{C}_9 \mathsf{C}_{12}^{-1}\right)^2, \\ & 2^{10}\, 4\sqrt{2}\, \mathsf{C}_4 \mathsf{C}_9. \end{aligned}$$

These terms are strictly less than 2^{171}, 2^{280}, and 2^{140}, respectively; the inverse factor uses $\mathsf{C}_{12} > 1$. Also $c_0 = \mathsf{C}_4/16 > 1$. These are bounds on the explicit theorem constants, not unspecified universal constants. An accompanying exact-rational checker evaluates the original formulas and verifies every displayed inequality.

Verification for the target and trained models. For the target, $\omega = (1, \sqrt{2})$ gives a fourth-order divisor minimum $\sqrt{2} - 1 > 2/5$. Its quartic coefficient norm is $\frac{3}{80}(1 + 1/\sqrt{2})^2 < 3/20$, while $\det A / \operatorname{tr} A = 1/48$. Thus $\lambda_{\min}(A) > 1/48 > 1/128$, $\|A\|_\infty < 1$, and $\|A^{-1}\|_\infty \le \sqrt{2}/\lambda_{\min}(A) < 96 < 128$. For each GC fit, the accompanying certificate checks (85)–(86) using exact rational arithmetic on both the exported decimal coefficients and the stored binary coefficients. The three quartic coefficient norms are approximately 0.142214, 0.130694, and 0.131082; the corresponding inverse-twist matrix norms are below 53. The check also covers the rounded quadratic frequencies in the exported artifacts: the same fourth-order divisor bound holds. The theorem therefore does not depend on interpreting a rounded frequency as an exact quadratic irrational.

The same exact-arithmetic checks also pass for all three GC models fitted at noise level 5×10^{-5}, for which the separate loss–Gram–twist transfer test passes. Their quartic coefficient norms are below 0.110, their inverse-twist norms are below 48, and their $\det A / \operatorname{tr} A$ lower bounds exceed 0.0208. For this quantitative amplitude statement, the additional direct coefficient check verifies the common bound $\lambda_{\min}(A) \ge 1/128$; the loss-based positivity test alone is not asserted to provide that stronger uniform margin for every seed.

At $a = 2^{-225}$, the enclosing balanced-coordinate radius is $6a \simeq 1.113 \times 10^{-67}$. This is a deliberately conservative sufficient bound, with all constants specified. It certifies the actual target and trained Hamiltonians on the annuli Ω_a; it neither asserts a torus fraction for the whole enclosing ball nor certifies the much larger simulation amplitudes. The canonical balancing $q_j = Q_j/\sqrt{\omega_j}$, $p_j = \sqrt{\omega_j} P_j$ maps these sets to the original modal variables without changing their relative Liouville measure.

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